OPEN AGENT HACKATHON 2026 · TRACK 2 · THE AGENT THAT CAN EXPLAIN WHY

# ExplainWhy

An agent that explains *why* your data sources disagree, and answers only with claims a critic has checked against the query results.

NVIDIA Nemotron | Zetaris data harness | Meterless World Model · H-MEM · Markovian

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github repo: see submission

THE PROBLEM

# "Why don't these numbers line up?" costs analysts hours

- A revenue series that just **stops** in 2018.
- A join that silently **drops** one company.
- Two figures for the same thing that **don't match**.

The cause is spread across sources: tagging changes, reference-data gaps, duplicated filings, real business events.

## A plain LLM is the wrong tool

It answers fluently, invents numbers, and mixes plausible causes with facts. You cannot tell which sentence came from the data.

**ExplainWhy works like a careful analyst:** hypotheses → checks → ruled out → only what the data shows, with receipts.

WHAT IT DOES

# Hypotheses, SQL evidence, a strict critic, a verified answer

**Recall**
H-MEM brings back prior verified findings, to re-check before use

→

**Plan**
3–5 competing hypotheses: data artifacts *and* real-world causes

→

**Investigate**
one bounded step at a time: read-only SQL in Zetaris, every result recorded as evidence E-xxxxxx

→

**Critique**
citations exist · numbers appear in cited rows · no vacuous zeros · LLM review

→

**Answer**
verified claims only; causes the data can't prove are labelled "likely"

↺ Unsupported claims go back to the investigator with the critic's reasons (up to 2 repair rounds). Hard limits: 8-minute run deadline, step budget with forced finish, capped retries, single-SELECT guard.

Every role runs on **NVIDIA Nemotron** (`nemotron-3-super-120b-a12b`). The full trace is the World Model event log: question, hypotheses, each query and its rows, each critic verdict, the answer.

REAL RUN · SAMPLE 2

# Why does IBM drop out of the SIC join?

```
$ python -m explainwhy --sample 2
[planner]  5 hypotheses: whitespace · zero-padding · code missing from reference · NULL · duplicate rows
[sql]      E-c38e8f  IBM's sic_code in company_profiles              → 1 row (3570)
[sql]      E-84b0e6  more than one row for IBM?                     → 1 row
[sql]      E-1943cf  sic_code = '3570' in sic_codes_table           → 0 rows
[sql]      E-a9171e  length before / after trim                     → 4 / 4
[sql]      E-be1c50  zero-padded variants ('03570')                 → 0 rows
[sql]      E-1943cf  repeated query refused
[critic]   round 1: "the whole table is zero-padded" PARTIAL → sent back for proof
[critic]   round 2: 5 of 5 claims SUPPORTED · hypotheses: 1 supported, 4 refuted

IBM's EDGAR profile lists sic_code 3570, but the company_dns SIC reference table contains no entry
for 3570 or any zero-padded variant, so the join drops IBM. This likely indicates 3570 is not in
the reference list (possibly obsolete).
  ✓ IBM's sic_code in edgar.company_profiles is 3570.                    [E-c38e8f]
  ✓ A search for '3570' and zero-padded variants returns zero rows.   [E-1943cf, E-be1c50]
  ✓ IBM's sic_code has no leading or trailing whitespace.                [E-a9171e]
```

Data: SEC EDGAR submissions + company_dns SIC list — two independent public sources

TRUST

# The critic decides what you get to read

## Deterministic checks (no model)

- Every claim cites evidence that was actually recorded.
- Every year and 4+-digit number in a claim appears in the cited rows or SQL.
- A zero result proves nothing if its filter value matches no rows (`entity_name = 'Apple'` when the table says "Apple Inc.").

## Then an LLM review

- Do the cited rows, and only those, show the claim? supported / partial / unsupported.
- Rejected claims are sent back with reasons; the investigator gathers more evidence or drops them.
- The final summary is written from verified claims only; a new number in it is rejected.

Caught during today's development runs: invented SIC codes (3599, 3500) not in any row, a "no rows" claim built on a misspelled company name, and partial claims sent back for proof.

SPONSOR TECHNOLOGY

# Where each sponsor sits in the build

| SPONSOR | HOW IT IS USED |
| --- | --- |
| **Zetaris** | The data harness. All evidence is Lightning SQL over the Zetaris REST API against views over live REST sources: SEC XBRL facts (5 REST tables unioned), SEC submissions, company_dns SIC list. Cross-source joins run in Zetaris. zetaris/01_create.sql · data.py |
| **Meterless** | **World Model**: event-sourced investigation state and trace. **H-MEM**: verified findings with source, provenance, confidence and audit ledger, recalled next time. **Markovian**: bounded steps with the COMPLETED / REMAINING / DECISIONS / CONTEXT carryover. engines/ |
| **NVIDIA** | Nemotron 3 Super 120B runs every role: planner, investigator, critic, writer. llm.py |
| **Cursor** | Not used: HackOS Cursor access was still "TBC" on 27 Oct. Built with Claude Code (allowed by the AI Usage Policy). |

RESULTS

# Three real anomalies in public data, explained with evidence

| QUESTION | VERIFIED ANSWER | RUN |
|---|---|---|
| Why does Apple's `Revenues` history stop at FY2018? | 0 rows after FY2018; the ASC 606 concept runs FY2019 on. ASC 606 adoption given as the *likely* cause. | 60 s · 3 steps |
| Why does IBM drop out of the SIC join? | IBM's SIC 3570 is absent from the reference list; whitespace, NULLs, zero-padding and duplicate rows ruled out. | 71 s · 9 steps · 2 critic rounds |
| Why does Tesla use two revenue concepts? | Only `Revenues` in FY2011–2017; both from FY2018, in the same filings, sometimes identical (duplicate tagging). | 47 s · 5 steps |

Markovian bounded context: 42–73 % fewer prompt tokens than re-sending the transcript, and the gap grows with run length. Offline replay of all three runs: `SAMPLE_MODE=true`.

# What is not done yet

- Recorded sample runs used the local mirror of the Zetaris views (same names, same SQL); the live Zetaris path is implemented against the documented REST API.
- The critic is strict on numbers and citations, but can still accept a weak interpretation worded as a fact.
- Small, bounded world: 3 companies, 2 revenue concepts.

## Next

- More Zetaris sources through its semantic layer (USL), so questions span many systems.
- H-MEM "dreaming": propose general rules from many verified findings, with human approval.
- A small web UI that shows each claim beside its rows.

TRY IT

# Runs in minutes, no GPU, standard-library Python

```
# offline replay of the recorded runs, no keys needed
SAMPLE_MODE=true python -m explainwhy --sample 1

# live NVIDIA Nemotron on the local data mirror
python -m explainwhy "Why does Tesla report revenue under two different XBRL concepts?"

# live Nemotron + live Zetaris (falls back to the mirror, and says so, if unreachable)
python -m explainwhy --zetaris --sample 2

# offline tests with a scripted model
python -m pytest -q
```

Each run writes report.md (answer, verified and dropped claims, hypotheses, every query with its rows), result.json, events.jsonl (the trace) and trace.log. All code was written on 27 Oct 2026; no pre-existing components.